

Decoding Web Memorization: A Semantic Membership Inference Attack on LLMs

Zhiyao Wu*
bc31072@um.edu.mo
University of Macau
Macau, China

Zi Liang*
zi1415926.liang@connect.polyu.hk
Hong Kong Polytechnic University
Hong Kong, China

Haibo Hu†
haibo.hu@polyu.edu.hk
Hong Kong Polytechnic University,
PolyU Research Centre for Privacy
and Security Technologies in Future
Smart Systems
Hong Kong, China

Abstract

While Large Language Models (LLMs) are increasingly deployed in Web applications such as search, dialogue, and recommendation systems, their reliance on large-scale Web data raises serious privacy concerns, particularly the risk of memorizing sensitive content. Existing Membership Inference Attacks (MIA) rely heavily on the surface form of inputs, rendering them ineffective against semantically preserved but structurally altered samples. This methodological weakness results in widespread false negatives and compromises the integrity of privacy evaluations in large-scale Web corpora. To address this limitation, we propose Adversarial Semantic Membership Inference Attack (ASMIA). ASMIA enhances MIA effectiveness by generating semantically diverse adversarial samples, extracting multi-layer attention features from the target model, and training a contrastive classifier that leverages similarity metrics and logarithmic probabilities to distinguish members from non-members. Experiments on LLMs trained with Wikipedia, a representative large-scale Web corpus, demonstrate that ASMIA significantly outperforms existing methods, highlighting the value of semantic perturbations and attention patterns in detecting training data leakage.

CCS Concepts

• Information systems → World Wide Web.

Keywords

Semantic Membership Inference, Adversarial Knowledge, Semantic Data, Web Memorization, Large Language Models

ACM Reference Format:

Zhiyao Wu, Zi Liang, and Haibo Hu. 2026. Decoding Web Memorization: A Semantic Membership Inference Attack on LLMs. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3774904.3792151>

*Equal contributions.

†Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. *WWW '26, Dubai, United Arab Emirates*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2307-0/2026/04

<https://doi.org/10.1145/3774904.3792151>

1 Introduction

Large Language Models (LLMs) have become indispensable tools in a wide range of decision-making tasks and are widely deployed in Web-driven applications such as search engines, conversational agents, content recommendation, and social media platforms [2, 7, 8, 31]. These models are trained on massive Web-scale corpora containing user-generated content, personal data, and publicly available documents. While LLMs exhibit strong generalization capabilities, they can also memorize and reproduce exact segments of their training data, creating significant privacy risks [18, 28]. For instance, in October 2025 Anthropic announced that conversations with its Claude chatbot would be collected for training by default unless users opted out [32]. This raised concerns that sensitive user dialogues could be memorized and later exposed, underscoring the urgency of developing robust membership inference techniques for Web-scale LLMs.

A key consequence of such memorization is the vulnerability to *Membership Inference Attacks* (MIA) [5, 11, 26, 30]. MIA aims to determine whether a particular sample was part of a model's training set. By exploiting differences in model behavior on training versus unseen data, MIA provides a systematic way to study how and under what conditions memorization occurs in large models. Under Web environments, this vulnerability is especially concerning: adversaries can craft targeted queries to extract sensitive information memorized by the model, thereby directly compromising user privacy and violating data protection regulations.

Traditional MIA typically relies on *surface-level model behaviors*, such as confidence score or output probability [10], in response to exact input samples. These methods often *assume access to inputs identical to those used during training* and perform well under such conditions. However, their effectiveness *significantly diminishes when the input is slightly altered* with paraphrasing, rewording, or formatting, even if the semantic meaning remains unchanged. This limitation is particularly critical in *real-world Web environments*, where user interactions are inherently noisy, diverse, and semantically flexible. For instance, attackers may query a model with *semantically equivalent but structurally different phrases* extracted from online discussions, making it unrealistic to assume perfect string matches. While corresponding research [24] attempts to mitigate such drawback with a simple rule-based lexical solution, they can only offer limited structural diversity and thus perform poorly on discriminating industrially variant training samples on LLMs. Consequently, it is vital to develop a membership inference algorithm that can robustly handle semantically equivalent yet

structurally diverse inputs, and thus reflect realistic Web-facing scenarios.

To address this issue, we propose ASMIA (Adversarial Semantic Membership Inference Attack), which introduces an adversarial generation strategy to systematically produce diverse, semantic-preserving variants. Unlike classical MIA, ASMIA actively searches for input transformations that maximize behavioral differences while maintaining semantic consistency, enabling a more fine-grained probing of model memorization and improving the ability to distinguish training from non-training samples. In doing so, ASMIA effectively generates semantic adversarial examples, designed to reveal memorized knowledge without altering input meaning, thereby enhancing model robustness analysis and interpretability [14, 37]. To operationalize this approach, ASMIA employs a fine-tuned LLM to generate multiple semantically consistent yet structurally diverse variants for each input, which are then filtered to ensure high semantic fidelity. The target LLM is queried with both original and adversarial inputs, and deep-layer attention representations are extracted to compute similarity scores across variants. These representation-based signals are combined with the model's output log-likelihood to form feature vectors, which are fed into a lightweight contrastive classifier for membership inference. By leveraging both internal representation shifts and output confidence, ASMIA achieves precise and robust detection of memorized content, uncovering privacy risks that remain hidden to traditional or rule-based semantic attacks, especially in Web-scale deployments. In summary, this paper makes the following contributions:

- We propose ASMIA, a novel adversarial framework that systematically probes semantic-level memorization in LLMs by generating semantically consistent yet structurally diverse inputs. To the best of our knowledge, this is the first work which considers semantic information in membership inference attacks.
- We implement ASMIA by integrating adversarial sample generation, attention-based analysis, and log-likelihood scoring. By leveraging attention stability and contrastive training, it effectively captures subtle membership signals beyond surface-level methods, offering interpretability into memorization behaviors that affect Web-scale systems.
- We evaluate ASMIA through extensive experiments on large language models with Web-scale training corpora. Compared with [24][29][35], the experimental results show that our ASMIA outperforms prior methods in distinguishing training data, providing more precise and interpretable insights into memorization and privacy risks, with direct implications for privacy-preserving deployment of LLMs in Web applications.

2 Related Work

Shokri et al. [30] first formalized the concept of MIA in machine learning. They demonstrated that even with black-box access to a model, an adversary could train a separate inference model to distinguish between training and non-training samples based on the target model's prediction behavior. In [4], Nicholas et al. analyzed

how overfitting and data influence contributed to membership inference attacks, showing that overfitting was sufficient but not necessary for privacy leakage. It highlighted that reducing memorization and controlling model outputs through regularization, confidence masking, or differentially private training can effectively mitigate membership inference risks. Carlini et al. [5] first demonstrated that large language models can memorize and reveal training data, enabling membership inference even with black-box access. Their work highlighted serious privacy risks as model size increases, laying the groundwork for MIA research in LLMs. Carlini et al. also introduced the likelihood ratio MIA for image classifiers and GPT-2 [3]. The core idea is to use a likelihood ratio test that compares confidence score distributions from shadow models trained with or without the target sample to infer membership. Following this line of work, Ma et al. [19] explored which components of LLMs are responsible for memorization and proposed pruning strategies to mitigate MIA risks. It indirectly related to MIA by analyzing memorization patterns in different layers and model components, providing insights into how memorization emerges during pre-training.

In [22], a non-parametric MIA was proposed that leverages embedding-based similarity to compare the neighborhood of a test sample with known training samples. This approach offers a lightweight and generalizable detection method without requiring access to model internals. Shi et al. [29] proposed Min-K to detect training data based on the rarity of low-probability words, requiring no reference model or prior corpus knowledge. Experiments on the WIKIMIA benchmark and real-world tasks (e.g., copyright and privacy auditing) showed significant improvements over prior detection methods. Building on limitations of Min-K, which relies on simple heuristics for pretraining data detection, Min-K++ [35] was introduced, a theoretically grounded method. It leveraged the insight that training examples tended to be local maxima under the model's learned distribution, translating detection into identifying high-probability modes. From a membership-inference perspective, Min-K++ provided a principled approach to determine which inputs the model has likely memorized, improving privacy auditing and mitigation. Building on Min-K++, SMIA [24] further enhanced membership inference by leveraging semantic content and input perturbations. Instead of relying solely on token probabilities, SMIA trained a neural network to analyze how the target LLM's outputs changed under semantically meaningful perturbations, capturing subtle differences between members and non-members. Similarly, SPV-MIA [20] detected membership by training a reference model on data generated from the target LLM, assuming these responses shared a similar structure with the original training set but remained disjoint due to model randomness. Membership was then inferred by comparing target outputs with reference outputs using a probabilistic variation metric. This approach reduced the need for manually collecting reference datasets, though its effectiveness depended on the similarity and diversity of the generated data relative to the true training set.

Building on prior membership inference methods, recent studies extended detection to larger units and diverse data types. Ren et al. [28] proposed dataset-level membership inference methods

that evaluate model memorization over larger sequences. By introducing paraphrased non-member suffixes following member prefixes, these methods can trigger and measure memorization effects without requiring ground-truth member or distribution-matched non-member data. Similarly, document-level MIA [23] assessed whether an LLM had seen an entire document during training. By leveraging model release dates and commonly used corpora to identify candidate member documents, this method demonstrated that membership could be reliably inferred even under black-box access. PREMIA [9] analyzed the vulnerability of DPO and PPO aligned LLMs to membership inference using a reference-based attack on preference data, showing that DPO models were more susceptible. Together, these studies illustrate that membership inference threats scale from individual tokens and samples to larger sequences, documents, and aligned preference data, motivating the need for robust and semantically-aware detection frameworks.

Summary: Overall, most existing membership inference methods focus on token or sample level detection. These approaches often rely on shallow probability signals or heuristic transformations, which limits their robustness, interpretability, and effectiveness in realistic Web-scale environments. To address these limitations, we propose ASMIA, a novel adversarial semantic membership inference framework that systematically probes semantic level memorization in LLMs by generating semantically consistent yet structurally diverse inputs. Unlike classical MIA, ASMIA actively searches for input transformations that maximize behavioral differences while preserving semantic consistency, enabling more fine-grained probing of model memorization and improving the ability to determine whether a sample belongs to the training set.

3 Design of ASMIA

3.1 Overview

In this work, we introduce **ASMIA**, an advanced MIA paradigm targeting LLMs. ASMIA determines whether a semantic concept, defined as a collection of semantically equivalent yet structurally diverse text sequences, was included in the model’s training data. For instance, “The company reported a loss last quarter” and “The firm experienced a financial downturn in the previous quarter” share the same semantic content despite differing in phrasing. By adopting an adversarial generation strategy, ASMIA creates diverse, meaning-preserving variants that challenge the model’s generation.

As shown in Figure 1, ASMIA consists of three main modules: **I)** Adversarial semantic sample generation module. We utilize a fine-tuned BART model to generate multiple semantically equivalent but structurally diverse variants for each input. To ensure quality, we apply semantic filtering using Sentence-BERT to retain high-similarity samples and structural filtering via edit distance to guarantee sufficient surface variation. **II)** Attention-based behavior analysis module. We input the original and adversarial samples into the target LLM (e.g., Qwen3-8B) and extract multi-layer attention weights. We compute the cosine similarity between attention vectors to quantify attention stability, under the hypothesis that in-distribution samples exhibit more consistent internal behavior. **III)** Contrastive membership inference module. We construct feature vectors using both attention similarity and model output log-likelihood. These are fed into a lightweight MLP classifier trained

with a hybrid loss combining cross-entropy and contrastive components. This allows the model to distinguish between members and non-members by learning both classification boundaries and representation closeness.

We will respectively detail these three modules in the following subsections.

3.2 Adversarial Semantic Sample Generation

In this module, we aim to construct semantically equivalent but structurally diverse variants of each input text, which are later used to probe the model’s membership behavior. We employ a fine-tuned BART [15, 25] model as the paraphrase generator due to its powerful sequence-to-sequence capabilities, which make it well-suited for generating fluent and semantically consistent paraphrases. BART combines the strengths of bidirectional and autoregressive transformers, enabling it to produce high-quality variations that preserve meaning while altering surface form. To further ensure the quality of generated samples, we introduce a two-stage filtering mechanism to guarantee both semantic fidelity and surface-level diversity of the generated samples.

The specific steps of the module are as follows:

Step 1: For each original input sample x_i , we use a BART-based paraphrasing model to generate a set of n candidate paraphrases:

$$\mathcal{X}_{cand}(x_i) = \{x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)}\}. \quad (1)$$

Step 2: We use a pre-trained Sentence-BERT model [27] to compute the cosine similarity between each candidate paraphrase $x_i^{(j)} \in \mathcal{X}_{cand}(x_i)$ and the original input x_i :

$$\text{sim}(x_i, x_i^{(j)}) = \cos(\text{SBERT}(x_i), \text{SBERT}(x_i^{(j)})). \quad (2)$$

Then, candidates with similarity below a threshold τ_{sem} are discarded, i.e.,

$$\mathcal{X}_{sem}(x_i) = \{x_i^{(j)} \in \mathcal{X}_{cand}(x_i) \mid \text{sim}(x_i, x_i^{(j)}) \geq \tau_{sem}\}. \quad (3)$$

The semantic similarity threshold $\tau_{sem} \in [0.85, 0.95]$ is typically set to 0.9 to ensure high semantic consistency [27]. This threshold balances retaining a sufficient number of paraphrased samples while preserving their original meanings. Users can freely adjust τ_{sem} to balance semantic fidelity and structural diversity. Specifically, increasing τ_{sem} enforces stricter semantic consistency, while decreasing τ_{sem} allows for greater structural variation.

Step 3: For structural diversity, we compute the normalized edit distance between each retained paraphrase and the original input:

$$d_e(x_i, x_i^{(j)}) = \frac{ED(x_i, x_i^{(j)})}{\max(|x_i|, |x_i^{(j)}|)}, \quad (4)$$

where $ED(\cdot, \cdot)$ denotes the Levenshtein edit distance. Only samples with d_e above a structural threshold τ_{str} are kept:

$$\mathcal{X}_{adv}(x_i) = \{x_i^{(j)} \in \mathcal{X}_{sem}(x_i) \mid d_e(x_i, x_i^{(j)}) \geq \tau_{str}\}. \quad (5)$$

The final output $\mathcal{X}_{adv}(x_i)$ is a high-quality adversarial semantic sample set for the original input x_i , which ensures semantic consistency while introducing sufficient surface-level variance for downstream probing. The structural threshold τ_{str} is set to 0.2 to ensure that generated samples exhibit significant surface-level variation. This value reflects a normalized edit distance of 20%, which

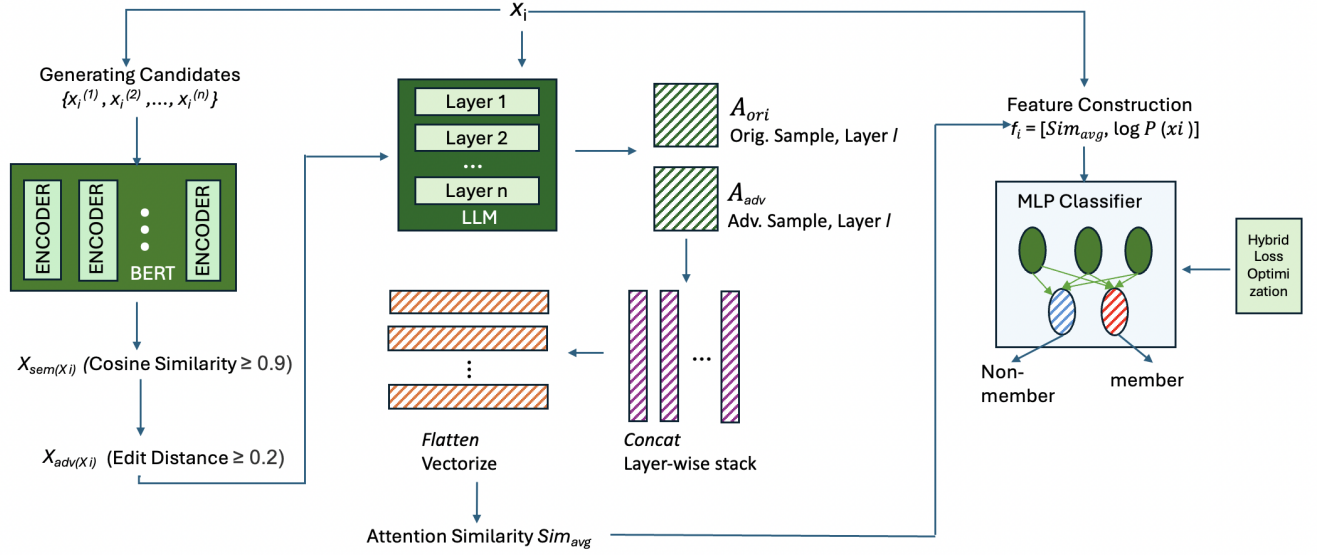


Figure 1: Architecture of ASMIA. For each input sample, semantically equivalent but structurally diverse variants are generated and filtered. The original and adversarial samples are then fed into the target LLM to extract and compare multi-layer attention weights, yielding an attention stability score. Finally, this score is combined with the model’s log-likelihood to form feature vectors for an MLP classifier trained with a hybrid cross-entropy and focal loss.

effectively filters out minor modifications such as small spelling changes or punctuation adjustments, while maintaining semantic equivalence.

Similar to τ_{sem} , users can freely adjust τ_{str} to balance structural variation against similarity to the original input. Specifically, increasing τ_{str} enforces more substantial surface-level changes, whereas decreasing τ_{str} yields paraphrases that remain closer in form to the original text.

3.3 Attention-based Behavior Analysis

In this module, we analyze the internal behavior of the target LLM by extracting and comparing multi-layer attention weights for the original and adversarial samples. Attention weights reflect how the model distributes its focus across different input tokens during processing, providing insight into the model’s internal decision-making process. By examining these attention patterns across different layers of the model, we aim to determine whether the model exhibits consistent behavior in response to semantically similar inputs.

Therefore, this process can be designed to quantify the stability of the model’s attention patterns, serving as a potential indicator of membership. The underlying intuition is that models tend to display more stable and predictable attention behavior for training samples they have memorized, whereas non-member inputs, particularly their adversarial variants, are more likely to provoke unstable or erratic responses [13, 21]. Consequently, attention consistency provides a fine-grained and structure-aware signal for distinguishing members from non-members, complementing and extending surface-level indicators such as logits or confidence scores.

The specific steps of the module are as follows:

Step 1: For each original input sample x_i and its adversarial sample set $\mathcal{X}_{adv}(x_i)$, we feed them into the target LLM (e.g., Qwen3-8B) configured to output attention weights from the last L layers:

$$\begin{aligned} A_{ori} &= \{A^{(l)}(x_i)\}_{l=1}^L, \\ A_{adv}^{(j)} &= \{A^{(l)}(x_i^{(j)'})\}_{l=1}^L, \quad x_i^{(j)'} \in \mathcal{X}_{adv}(x_i), \end{aligned} \quad (6)$$

where $A^{(l)}(\cdot)$ represents the attention matrices extracted from the l -th layer.

Step 2: We flatten and concatenate the attention matrices from the L layers into vectors:

$$\begin{aligned} v_{ori} &= \text{flatten}(\text{concat}(A_{ori})), \\ v_{adv}^{(j)} &= \text{flatten}(\text{concat}(A_{adv}^{(j)})), \end{aligned} \quad (7)$$

where the $\text{concat}(\cdot)$ function combines attention matrices from all layers into a single tensor, and $\text{flatten}(\cdot)$ aims to reshape this tensor into a one-dimensional vector for downstream comparison. This process allows us to capture and compare the full spectrum of attention behavior across layers.

Step 3: To measure attention stability, we compute the cosine similarity between the original and each adversarial attention vector:

$$\text{sim}^{(j)} = \cos(v_{ori}, v_{adv}^{(j)}), \quad (8)$$

and then calculate the average similarity over all adversarial samples:

$$\text{Sim}_{avg} = \frac{1}{|\mathcal{X}_{adv}(x_i)|} \sum_j \text{sim}^{(j)}. \quad (9)$$

The average similarity score Sim_{avg} thus serves as an indicator of membership: a higher value suggests that the model reacts to

Algorithm 1 Adversarial Semantic Membership Inference Attack

```

1: Input: Query sample  $x_i$ , candidate number  $n$ , paraphrasing
   model  $f_p$ , sentence embedding model  $f_s$ , similarity threshold
    $\tau_{sem}$ , structure threshold  $\tau_{str}$ , targeted LLM  $f_t$ , monitor layer
   number  $L$ , and MIA classifier  $f_c$ .
2: Return:  $\hat{y}_i \in [0, 1]$  which denotes the confidence that  $x_i$  is
   a training member of  $f_t$  under the discrimination of ASMIA
   algorithm.
3: // Stage 1: Adversarial Semantic Sample Generation.
4: // Initialize.
5:  $n_i \leftarrow 1, n_j \leftarrow 1, \mathcal{X}_{cand} \leftarrow \emptyset, \mathcal{X}_{adv} \leftarrow \emptyset;$ 
6: while  $n_i \leq n$  do
7:    $x_i^j \leftarrow f_p(x_i);$ 
8:    $\mathcal{X}_{cand} \leftarrow \mathcal{X}_{cand} \cup \{x_i^j\};$ 
9:    $n_i \leftarrow n_i + 1;$ 
10: end while
11: // Filtering.
12: while  $n_j \leq n$  do
13:    $x_i^j \leftarrow \text{get}(\mathcal{X}_{cand}, n_j);$ 
14:    $e_i \leftarrow f_s(x_i), e_j \leftarrow f_s(x_i^j);$ 
15:    $s_j \leftarrow \cos(e_i, e_j);$ 
16:    $l_j \leftarrow ED(x_i, x_i^j) / \max(|x_i|, |x_i^j|);$ 
17:   if  $s_j \geq \tau_{sem} \ \&\& \ l_j \geq \tau_{str}$  then
18:      $\mathcal{X}_{adv} \leftarrow \mathcal{X}_{adv} \cup \{x_i^j\};$ 
19:   end if
20:    $n_j \leftarrow n_j + 1;$ 
21: end while
22: // Stage 2: Attention Based Analysis.
23:  $sim_{avg} \leftarrow 0, A_{ori} \leftarrow \emptyset, A_{adv} \leftarrow \emptyset, l \leftarrow 1;$ 
24: while  $l \leq L$  do
25:    $A^{(l)} \leftarrow \text{getAttentions}(x_i, f_t);$ 
26:    $A_{ori} \leftarrow A_{ori} \cup \{A^{(l)}\};$ 
27:    $l \leftarrow l + 1;$ 
28: end while
29:  $v_{ori} \leftarrow \text{flatten}(\text{concat}(A_{ori}));$ 
30: for  $x_i^j \in \mathcal{X}_{adv}$  do
31:    $A_{adv}^j \leftarrow \emptyset, l \leftarrow 1;$ 
32:   while  $l \leq L$  do
33:      $A^{(l)} \leftarrow \text{getAttentions}(x_i^j, f_t);$ 
34:      $\mathcal{A}_{adv}^j \leftarrow \mathcal{A}_{adv}^j \cup \{A^{(l)}\};$ 
35:      $l \leftarrow l + 1;$ 
36:   end while
37:    $v_{ori} \leftarrow \text{flatten}(\text{concat}(A_{adv}^j));$ 
38:    $sim_j \leftarrow \cos(v_{ori}, v_{adv}^j);$ 
39:    $sim_{avg} \leftarrow sim_{avg} + sim_j;$ 
40: end for
41:  $sim_{avg} \leftarrow sim_{avg} / |\mathcal{X}_{adv}|;$ 
42: // Stage 3: Membership Inference.
43:  $p_i \leftarrow \log(\text{Probabilities}(f_t, x_i));$ 
44:  $f_i \leftarrow \text{concat}(sim_{avg}, p_i);$ 
45:  $y_i \leftarrow f_c(f_i);$ 
46: return  $y_i$ 

```

the adversarial variants in a stable and predictable manner, which

is more likely for training (member) samples. In contrast, a lower Sim_{avg} indicates less stable attention responses, suggesting that the input is more likely to be a non-member. Consequently, we adopt Sim_{avg} as a key feature for subsequent membership inference classification.

3.4 Contrastive Membership Inference

In this module, we construct feature vectors by combining the attention similarity metric Sim_{avg} obtained from the previous module with the model output log-likelihood $\log P(x_i)$ for each input sample x_i . Here, $\log P(x_i)$ is calculated as the sum of the log probabilities of the tokens in x_i predicted by the target model, reflecting the model's confidence in generating the sample. These combined features serve as input to a lightweight Multi-Layer Perceptron (MLP) classifier that predicts membership probabilities.

The specific steps of the module are as follows:

Step 1: For each input sample x_i , form the feature vector:

$$f_i = [Sim_{avg}, \log P(x_i)], \quad (10)$$

where Sim_{avg} measures attention stability and $\log P(x_i)$ denotes the log-likelihood score assigned by the target LLM. This step constructs feature vectors that capture both attention stability and model confidence, providing complementary indicators of membership.

Step 2: The feature vectors f_i are fed into a 2-layer MLP classifier with ReLU activations, which outputs a membership probability $\hat{y}_i \in [0, 1]$.

Step 3: The classifier is trained using a hybrid loss function combining cross-entropy loss [36] $\mathcal{L}_{CE}$ to optimize classification accuracy and focal loss $\mathcal{L}_{FL}$ to encourage representation clustering of members and separation of non-members:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{FL}, \quad (11)$$

where λ is a weighting hyperparameter.

The cross-entropy loss $\mathcal{L}_{CE}$ measures the discrepancy between predicted membership probabilities and true labels. For binary classification, given a true label $y \in \{0, 1\}$ and predicted probability $\hat{y} \in [0, 1]$, it is defined as:

$$\mathcal{L}_{CE} = -(y \log \hat{y} + (1 - y) \log(1 - \hat{y})). \quad (12)$$

To address class imbalance in membership inference, we incorporate focal loss into the loss function framework. The key formulations are as follows:

First, we define the adjusted probability p_t based on the true label y :

$$p_t = \begin{cases} \hat{y} & \text{if } y = 1, \\ 1 - \hat{y} & \text{if } y \neq 1 \end{cases} \quad (13)$$

Then, the focal loss [16, 17] is calculated as:

$$\mathcal{L}_{FL} = -\alpha \cdot (1 - p_t)^\gamma \cdot \log(p_t + \epsilon). \quad (14)$$

In our implementations, we carefully tune hyperparameters to ensure the focal loss effectively addresses class imbalance. We set the balancing factor $\alpha = 0.25$ to mitigate bias toward majority classes, the focusing parameter $\gamma = 2.0$ to amplify loss for hard-to-classify samples, and introduce a small constant ϵ to avoid numerical instability from $\log(0)$ calculations. Here, N denotes the total number of samples in each batch used for loss computation.

The goal of these three steps is to build a robust membership inference classifier that leverages both the internal behavioral signals and output-based confidence measures of the target model.

4 Experiments

4.1 Experimental Setup

Datasets. We conduct experiments on the WikiMIA datasets, which serve as a benchmark for evaluating membership inference attack methods. WikiMIA [29, 33] is particularly suitable for WWW or web-related applications because it is derived from Wikipedia, a large-scale, openly editable web knowledge base. The dataset reflects typical web content characteristics, including diverse topics, varying writing styles, and natural linguistic structures, which closely resemble the textual data commonly found in online platforms and web services. Each dataset is divided into member and non-member samples, providing a controlled setting to assess an adversary’s ability to distinguish training data from unseen inputs, thereby enabling realistic evaluation of MIA methods in web scenarios.

Models. We select Mistral-7B, Qwen3-8B, and Pythia-12B for our experiments as representative open-source LLMs spanning lightweight, multilingual, and large-scale configurations, which enables a comprehensive evaluation of ASMLIA. Mistral-7B is lightweight and suitable for fast prototyping and method validation. Qwen3-8B supports multilingual input and facilitates cross-lingual evaluation. Pythia-12B features a large-scale architecture and publicly available training data, making it particularly well-suited for membership inference assessment and the analysis of semantic memorization.

Baselines. We evaluate the performance of ASMLIA in comparison with traditional MIA methods, including Zlib [6], LOSS [4], Min-K [29] and Min-K++ [35], as well as the semantic-level MIA method SMIA [24]. Zlib preserves equivalence in model responses and applies constraint-based regularization, reducing leakage signals and mitigating membership inference risks. LOSS regularization mitigates privacy risks by reducing generalization gaps and constraining attribute influence, thereby resisting membership and attribute inference attacks. Min-K detects training data members via token-level probabilities but may miss deeper semantic memorization. Min-K++ improves accuracy and robustness by interpreting training samples as local maxima in the model’s learned distribution, yet still relies on surface-level signals. SMIA leverages semantic content and input perturbations to capture memorization even for paraphrased inputs, but it does not use adversarial transformations. In contrast, ASMLIA combines semantic-preserving transformations with adversarial examples, effectively revealing subtle membership signals overlooked by previous methods and achieving superior performance across diverse LLMs.

Metric. We use the following **metrics** to evaluate our ASMLIA:

- **AUC** (Area Under the ROC Curve): AUC Measures the area under the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at different thresholds. It reflects the model’s overall ability to distinguish between member and non-member samples. A value of 1 indicates perfect separation, while 0.5 corresponds to random guessing.

- **ACC:** ACC measures the proportion of correctly classified samples over all samples:

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}},$$

where TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives.

- **TPR@5%FPR** (True Positive Rate at 5% False Positive Rate): The true positive rate when the false positive rate is fixed at 5%. It measures how effectively the attack identifies members while maintaining a low false positive rate:

$$\text{TPR@5\%FPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

Our source code is available at <https://anonymous.4open.science/r/Decoding-Web-Memorization-23E3/>.

4.2 Experimental Results

Overall Performance Comparison. Table 1 shows that ASMLIA consistently outperforms all baseline methods across Mistral-7B [12], Qwen3-8B [34], and Pythia-12B [1], demonstrating its superior ability to capture semantic-level membership signals in web-adapted scenarios. Zlib preserves equivalence in model responses and applies constraint-based regularization, but primarily targets low-level output consistency, often missing subtle memorization traces. LOSS reduces generalization gaps and controls attribute influence, yet its effectiveness diminishes when models memorize semantic information across diverse contexts. Min-K and Min-K++ analyze token probabilities and local maxima in the learned distribution, improving detection accuracy and robustness, but rely largely on surface-level signals and cannot fully exploit semantic cues. SMIA advances to semantic-level detection through input perturbations, yet it does not incorporate adversarial transformations to reveal deeper membership patterns. In contrast, ASMLIA combines semantic-preserving transformations with adversarial examples, effectively exposing subtle and latent membership signals overlooked by prior methods. The results in Table 1 clearly illustrate that ASMLIA achieves consistently higher detection performance across models and configurations, validating its effectiveness for membership inference in large language models. Specifically, Figure 2 visualizes the ACC comparison in different methods. WT represents WikiMIA test data, i.e. non-members are sampled from the same data distribution as member.

Figure 3 further quantifies the performance of all methods under a strict false positive constraint (5% FPR), which is particularly critical in web scenarios where mislabeling non-member content (e.g., user-generated comments or public web snippets) must be minimized to maintain trust and compliance. As shown in Figure 3, ASMLIA consistently achieves the highest TPR@5%FPR across all three models. For large-scale models such as Pythia-12B, ASMLIA attains nearly twice the TPR@5%FPR of semantic-level baselines and more than three times that of traditional methods. These results highlight ASMLIA’s unique capability to simultaneously ensure high member identification accuracy and effective false positive control, which is essential for practical deployment in web applications.

ROC Curve Analysis. Figure 4 presents the ROC curves for all methods. However, we only emphasize on comparing three representative methods in this section (ASMLIA, SMIA, and Min-K++),

Table 1: Overall Performance Comparison. Table 1 compares ASMIA with baseline methods on WT data. ASMIA achieves the best performance across all models, showing clear gains in AUC and TPR@5%FPR, which demonstrates its superior discriminative ability and robust generalization.

Method	Mistral-7B			Qwen3-8B			Pythia-12B		
	AUC(%)	ACC(%)	TPR@5%FPR(%)	AUC(%)	ACC(%)	TPR@5%FPR(%)	AUC(%)	ACC(%)	TPR@5%FPR(%)
LOSS	48.10	52.95	4.93	48.10	52.58	4.93	55.13	55.90	8.80
Zlib	48.78	53.32	2.11	47.73	53.14	8.45	51.26	54.24	2.11
Min-K (10%)	50.46	53.51	6.34	48.95	52.77	7.75	56.80	56.83	10.56
Min-K (20%)	52.44	54.06	4.93	48.14	52.58	5.28	58.67	57.93	10.56
Min-K (30%)	51.95	54.06	7.04	46.23	52.58	4.23	56.23	55.90	4.58
Min-K++ (10%)	54.56	55.35	6.34	49.68	53.14	3.52	61.12	58.30	6.34
Min-K++ (20%)	54.13	55.90	4.23	50.69	53.14	6.34	61.75	59.41	8.80
Min-K++ (30%)	53.56	55.54	5.99	50.53	52.95	8.10	62.90	60.33	11.97
SMIA	68.92	64.20	10.56	62.77	59.23	11.97	68.86	63.47	11.27
ASMIA	72.17	64.40	21.41	64.51	59.54	15.64	76.23	70.11	27.83

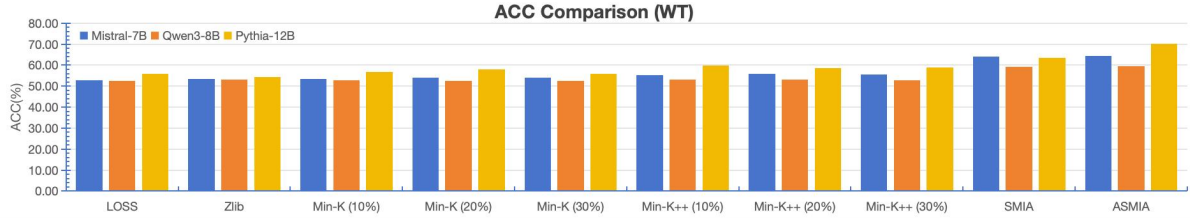


Figure 2: ACC Comparison.

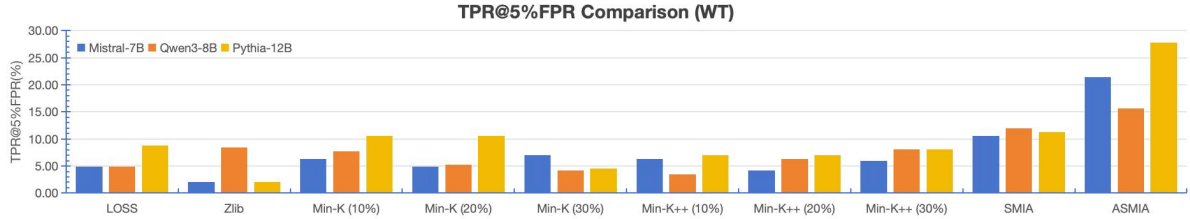


Figure 3: TPR@5%FPR Comparison.

selected due to space limitations. As shown in Figure 4 (a), ASMIA achieves ROC curves that are consistently above the random guessing line ($AUC = 0.5$) and display a steep upward trend, particularly at low FPR values. This demonstrates ASMIA’s ability to effectively identify member samples while maintaining strict false positive control, which is crucial for web scenarios where misidentifying non-member content could undermine user trust or violate data regulations. The steepest trend appears in the large-scale model, supporting the hypothesis that larger models retain finer-grained memorization that ASMIA’s adversarial semantic perturbations can effectively expose. In contrast, Figure 4 (b) shows that SMIA exhibits weaker discriminative power, with flatter ROC curves and substantially lower TPR at low FPR levels, confirming that its rule-based perturbations lack sufficient structural diversity to fully reveal memorization effects, particularly in multilingual (Qwen3-8B) and large-scale (Pythia-12B) models. Finally, as illustrated in Figure 4

(c), Min-K++ (10%) performs best among traditional MIA baselines but still remains close to the random guessing line across all models, far behind semantic-level methods. This limitation stems from traditional methods’ reliance on surface-form signals such as output probability or compression loss, which cannot adequately handle the coexistence of semantic equivalence and structural variation in web text, leading to frequent misclassification.

Model-Scale Sensitivity Analysis. Combined with Table 1 and ROC curves 4, ASMIA demonstrates strong robustness across models of different scales. For the lightweight model (Mistral-7B), ASMIA maintains the highest performance among all methods. Even with limited model capacity, which reduces the ability to store detailed web text, ASMIA’s attention-based feature analysis can still capture subtle membership signals and outperform baseline approaches. For the multilingual model (Qwen3-8B), ASMIA continues to outperform SMIA. Unlike SMIA, which struggles with

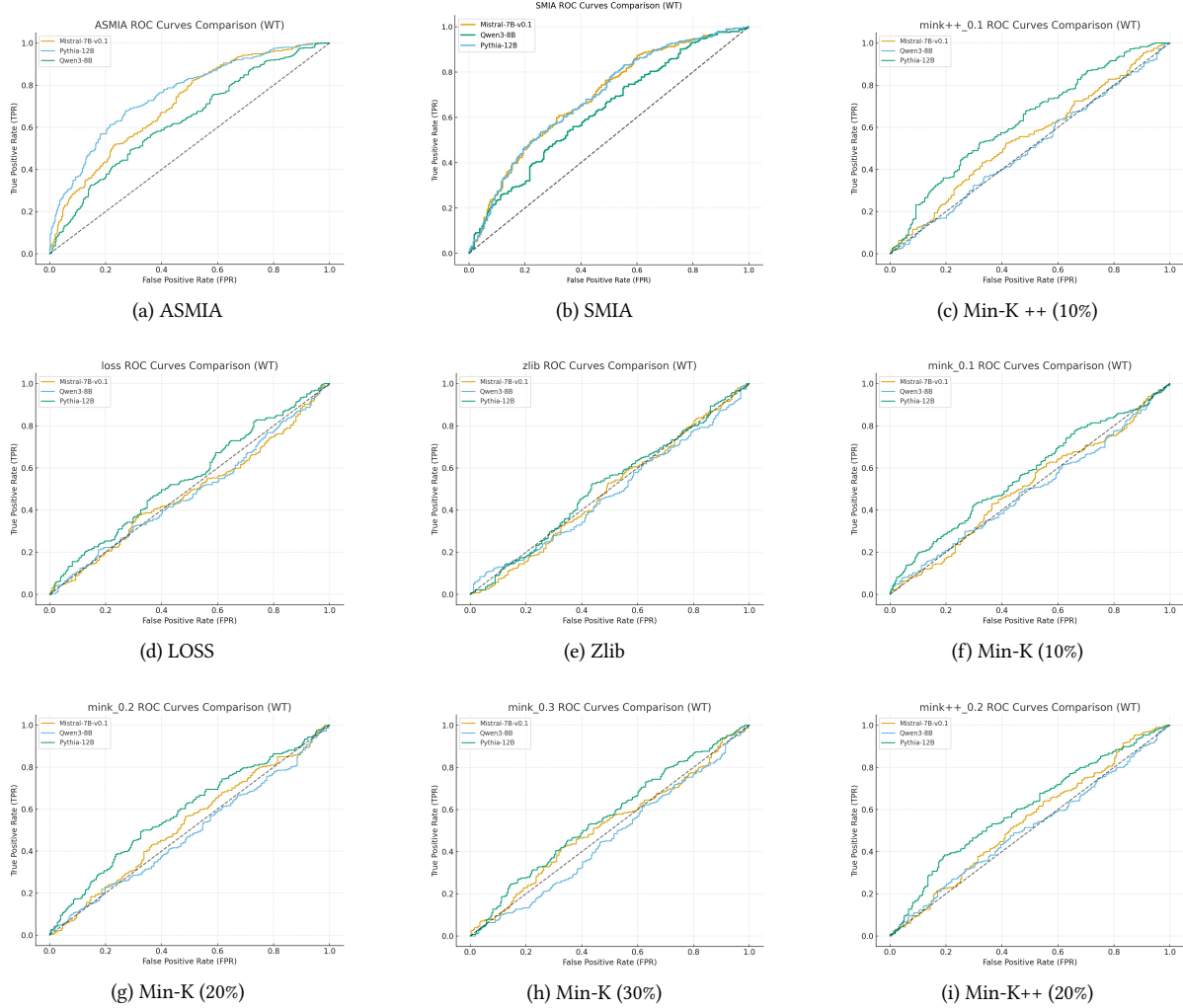


Figure 4: ROC Performance Comparison for All Methods.

multilingual semantic adaptation, ASMLA’s adversarial semantic generation preserves cross-lingual semantic consistency, making it suitable for multilingual web scenarios such as cross-lingual web search and multilingual social media. For the large-scale model (Pythia-12B), ASMLA remains superior. As model scale increases and the ability to memorize granular web text strengthens, ASMLA’s adversarial perturbations and contrastive classification better exploit fine-grained memorization traces, thereby avoiding the performance saturation observed in traditional methods and SMIA.

5 Conclusion

In this work, we proposed ASMLA, an adversarial semantic membership inference attack framework that systematically probes semantic-level memorization in large language models. By generating semantically consistent yet structurally diverse variants and leveraging both attention-based internal representations and output log-likelihoods, ASMLA effectively identifies training data

members beyond the capabilities of traditional or rule-based semantic attacks. Extensive experiments demonstrate its superior performance and interpretability, revealing privacy risks in Web-scale LLM deployments.

For future work, we plan to explore algorithmic enhancements for ASMLA to improve efficiency and effectiveness, including optimized adversarial input generation and more scalable attention-based representation extraction for large models.

6 Acknowledgments

We sincerely thank the reviewers for their detailed suggestions. This work was supported by the Ministry of Science and Technology of the People’s Republic of China (Grant No: 2025YFE0200100), the Research Grants Council (Grant No: 15210023, and 15207725), and the Innovation and Technology Fund (Grant No: ITS-140-23FP), Hong Kong SAR, China.

References

- [1] Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, et al. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*. PMLR, Cambridge, MA, USA, 2397–2430.
- [2] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, and et al. 2020. Language Models are Few-Shot Learners. In *Advances in Neural Information Processing Systems*, Vol. 33. Curran Associates, Inc., New Orleans, USA, 1877–1901.
- [3] Nicholas Carlini, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramer. 2022. Membership inference attacks from first principles. In *IEEE symposium on security and privacy (SP)*. IEEE, San Francisco, USA, 1897–1914.
- [4] Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfr Erlingsson, et al. 2018. Privacy Risk in Machine Learning: Analyzing the Connection to Overfitting. In *IEEE 31st Computer Security Foundations Symposium*. IEEE, Oxford, UK, 268–282.
- [5] Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfr Erlingsson, et al. 2021. Extracting training data from large language models. In *30th USENIX security symposium*. USENIX Association, Online, 2633–2650.
- [6] Sarah Cohen, Werner Nutt, and Yehoshua Sagie. 2007. Deciding equivalences among conjunctive aggregate queries. *J. ACM* 54, 2, Article 5 (April 2007), 50 pages. doi:10.1145/1219092.1219093
- [7] Yang Deng, An Zhang, Yankai Lin, Xu Chen, Ji-Rong Wen, and Tat-Seng Chua. 2024. Large language model powered agents in the web. In *Companion Proceedings of the ACM Web Conference (WWW)*. ACM, Singapore, 1242–1245.
- [8] Paul Duetting, Vahab Mirrokni, Renato Paes Leme, Haifeng Xu, and Song Zuo. 2024. Mechanism design for large language models. In *Proceedings of the ACM Web Conference (WWW)*. ACM, Singapore, 144–155.
- [9] Qizhang Feng, Siva Rajesh Kasa, SANTHOSH KUMAR KASA, Hyokun Yun, Choon Hui Teo, and Sravan Babu Bodapati. 2025. Exposing Privacy Gaps: Membership Inference Attack on Preference Data for LLM Alignment. In *International Conference on Artificial Intelligence and Statistics*. PMLR, Mai Khao, Thailand, 5221–5229.
- [10] Dominik Hintersdorf, Lukas Struppek, and Kristian Kersting. 2022. To trust or not to trust prediction scores for membership inference attacks. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI)*. IJCAI, Vienna, Austria, 3043–3049.
- [11] Li Hu, Anli Yan, Hongyang Yan, Jin Li, Teng Huang, Yingying Zhang, Changyu Dong, and Chunsheng Yang. 2023. Defenses to membership inference attacks: A survey. *Comput. Surveys* 56, 4 (Nov. 2023), 1–34. doi:10.1145/3620667
- [12] Albert Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Deven-dra Singh Chiplot, and et al. 2023. Mistral 7B. Retrieved Sep 27, 2023 from <https://mistral.ai/news/announcing-mistral-7b>
- [13] Tian Lan, Jinyuan Xu, Xue He, Jenq-Neng Hwang, and Lei Li. 2025. Attention Consistency for LLMs Explanation. arXiv:2509.17178 doi:10.48550/arXiv.2509.17178
- [14] Yuan-Heng Lee, Josh Jia-Ching Ying, and Vincent S Tseng. 2025. HaGAR: Hardness-aware Generative Adversarial Recommender. In *Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining*. ACM, Hannover, Germany, 838–846.
- [15] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. 2020. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Online, 7871–7880.
- [16] Xiang Li, Chengqi Lv, Wenhai Wang, Gang Li, Lingfeng Yang, and Jian Yang. 2023. Generalized Focal Loss: Towards Efficient Representation Learning for Dense Object Detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 3 (March 2023), 3139–3153. doi:10.1109/TPAMI.2022.3180392
- [17] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision ICCV*. IEEE Computer Society, Venice, Italy, 2999–3007.
- [18] Mingrui Liu, Sixiao Zhang, and Cheng Long. 2025. Mask-based Membership Inference Attacks for Retrieval-Augmented Generation. In *Proceedings of the ACM Web Conference (WWW)*. ACM, Sydney, Australia, 2894–2907.
- [19] Xinyin Ma, Gongfan Fang, and Xinchao Wang. 2023. LLM-Pruner: On the Structural Pruning of Large Language Models. In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems (NeurIPS 2023)*. Curran Associates, Inc., New Orleans, USA, 1–10.
- [20] Xinyin Ma, Gongfan Fang, and Xinchao Wang. 2024. Membership Inference Attacks against Fine-tuned Large Language Models via Self-prompt Calibration. In *Advances in Neural Information Processing Systems*, Vol. 37. Curran Associates, Inc., Vancouver, Canada, 134981–135010.
- [21] Disha Makhija, Manoj Ghuhan Arivazhagan, Vinayshekhar Bannihatti Kumar, and Rashmi Gangadharaiyah. 2025. Neural Breadcrumbs: Membership Inference Attacks on LLMs Through Hidden State and Attention Pattern Analysis. arXiv:2509.05449 doi:10.48550/arXiv.2509.05449
- [22] Justus Mattern, Fatemehsadat Miresghallah, Zhijiang Jin, Bernhard Schoelkopf, Mrinmaya Sachan, and Taylor Berg-Kirkpatrick. 2023. Membership Inference Attacks against Language Models via Neighbourhood Comparison. In *The 61st Annual Meeting Of The Association For Computational Linguistics*. Association for Computational Linguistics, Toronto, Canada, 11330–11343.
- [23] Matthieu Meeus, Shubham Jain, Marek Rei, and Yves-Alexandre de Montjoye. 2024. Did the Neurons Read your Book? Document-level Membership Inference for Large Language Models. In *33rd USENIX Security Symposium*. USENIX Association, Philadelphia, USA, 2369–2385.
- [24] Hamid Mozaffari and Virendra J Marathe. 2024. Semantic membership inference attack against large language models. arXiv:2406.10218 doi:10.48550/arXiv.2406.10218
- [25] Yuya Ogasa, Tomoyuki Kajiwara, and Yuki Arase. 2024. Controllable paraphrase generation for semantic and lexical similarities. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. ELRA and ICCL, Torino, Italy, 3927–3942.
- [26] Haritz Puerto, Martin Gubri, Sangdo Yun, and Seong Joon Oh. 2025. Scaling Up Membership Inference: When and How Attacks Succeed on Large Language Models. In *Findings of the Association for Computational Linguistics (NAACL)*. Association for Computational Linguistics, Albuquerque, USA, 4165–4182.
- [27] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Association for Computational Linguistics, Hong Kong, China, 3982–3992.
- [28] Jie Ren, Kangrui Chen, Chen Chen, Vikash Sehwal, Yue Xing, Jiliang Tang, and Lingjuan Lyu. 2025. Self-comparison for dataset-level membership inference in large (vision-) language model. In *Proceedings of the ACM on Web Conference (WWW)*. ACM, Sydney, Australia, 910–920.
- [29] Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. 2024. Detecting pretraining data from large language models. In *12th International Conference on Learning Representations (ICLR)*, Vol. 37. OpenReview, Vienna, Austria, 1–18.
- [30] Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. 2017. Membership inference attacks against machine learning models. In *2017 IEEE symposium on security and privacy (SP)*. IEEE, San Jose, CA, 422–431.
- [31] Yulong Wang, Hong Li, and Ni Wei. 2025. SAHSD: Enhancing Hate Speech Detection in LLM-Powered Web Applications via Sentiment Analysis and Few-Shot Learning. In *Proceedings of the ACM Web Conference (WWW)*. ACM, Sydney, Australia, 3014–3025.
- [32] Wired. 2025. Anthropic Will Use Claude Chats for Training by Default. Retrieved October 1, 2025 from <https://www.wired.com/story/anthropic-using-claude-chats-for-training-how-to-opt-out>
- [33] Roy Xie, Junlin Wang, Ruomin Huang, Minxing Zhang, Rong Ge, Jian Pei, Neil Gong, and Bhuwan Dhingra. 2024. ReCaLL: Membership Inference via Relative Conditional Log-Likelihoods. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Miami, USA, 8671–8689.
- [34] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 technical report. arXiv:2505.09388 doi:10.48550/arXiv.2505.09388
- [35] Jingyang Zhang, Jingwei Sun, Eric Yeats, Yang Ouyang, Martin Kuo, Jianyi Zhang, Hao Frank Yang, and Hai Li. 2024. Min-k%+: Improved baseline for detecting pre-training data from large language models. arXiv:2404.02936 doi:10.48550/arXiv.2404.02936
- [36] Zhilu Zhang and Mert Sabuncu. 2018. Generalized Cross Entropy Loss for Training Deep Neural Networks with Noisy Labels. In *Advances in Neural Information Processing Systems 31*. NeurIPS, Montreal, Canada, 8792–8802.
- [37] Ziming Zhao, Zhaoxuan Li, Tingting Li, Jiongchi Yu, Fan Zhang, and Rui Zhang. 2024. Improving model robustness against adversarial examples with redundant fully connected layer. In *Companion Proceedings of the ACM Web Conference 2024*. ACM, Singapore, 529–532.